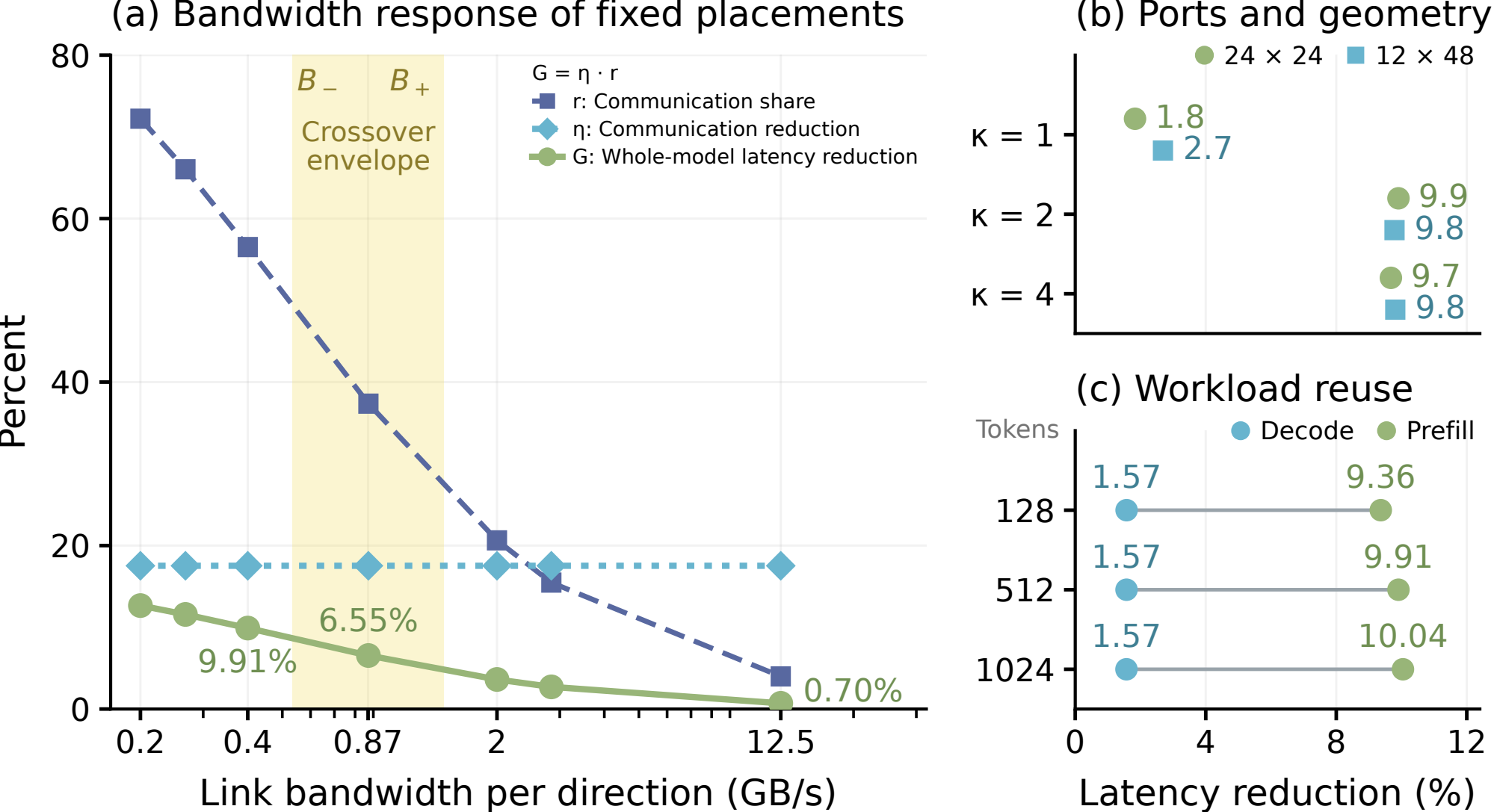
(a) Bandwidth response of fixed placements
Percent
80
60
40
20
0
B_-
B_+
Crossover envelope
G = η · r
r: Communication share
η: Communication reduction
G: Whole-model latency reduction
9.91%
6.55%
0.70%
0.2
0.4
0.87
2
12.5
Link bandwidth per direction (GB/s)
(b) Ports and geometry
24 × 24
12 × 48
κ = 1
κ = 2
κ = 4
1.8
2.7
9.9
9.8
9.7
9.8
(c) Workload reuse
Tokens
Decode
Prefill
128
512
1024
1.57
9.36
1.57
9.91
1.57
10.04
0
4
8
12
Latency reduction (%)